

Safety-oriented pre-event severity prediction of lithium-ion battery thermal runaway using interpretable machine learning

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ABSTRACT

Thermal runaway is one of the most critical failure modes of lithium-ion batteries and poses major safety challenges for electric vehicles and energy storage systems. Existing studies have primarily focused on trigger mechanisms and post-event characterization, whereas practical methods for assessing thermal runaway consequence severity before event occurrence remain limited. To address this gap, this study proposes an interpretable machine learning framework for pre-event severity prediction using only structural and engineering metadata of lithium-ion batteries. A hazard-oriented Severity Index is constructed by integrating normalized indicators of heat release and material ejection, and thermal runaway events are further classified into Low, Medium, and High severity levels. A CatBoost-based classifier is then developed to handle heterogeneous battery metadata and predict severity categories. The model attains an overall classification accuracy of 81% and successfully identifies all High severity events in the test set (recall = 1.00). SHAP analysis further reveals the key metadata features governing severity prediction. The proposed framework provides a lightweight and practical tool for early-stage battery safety screening, test prioritization, and proactive risk-oriented decision-making.

1. Introduction

Lithium-ion batteries are extensively utilized in electric vehicles and large-scale energy storage systems owing to their high energy density and advantageous electrochemical properties (Yang et al., 2026; Kul-karni et al., 2026). However, the ongoing increase in cell-level energy density and system integration has heightened critical safety concerns. Among various failure modes, thermal runaway is considered the most severe and hazardous, frequently characterized by intense heat release, gas emission, and material ejection (Mahdy et al., 2025; Dai and Panahi, 2025; Liu et al., 2025). These phenomena can escalate from localized cell failures to fires, explosions, or cascading system-level accidents, thereby posing significant risks to personnel safety, equipment integrity, and operational continuity. From a process safety standpoint, understanding and managing the consequences of thermal runaway constitute a primary challenge in battery safety engineering (Conzen et al., 2023; Zhang et al., 2024a).

Extensive experimental investigations have been conducted to

characterize thermal runaway behavior and its associated consequences (Musso et al., 2025; Chai et al., 2024). Calorimetric measurements (Sharp et al., 2022), gas analysis (Wang et al., 2025a), and ejected mass characterization (Willstrand et al., 2025) are commonly employed to quantify heat release intensity and material ejection during thermal runaway events. These post-event assessment methods have yielded valuable insights into failure mechanisms and hazard characteristics, contributing to the development of safety standards and mitigation strategies. However, such approaches are inherently resource-intensive, necessitate complex experimental setups, and are limited to application after the full development of thermal runaway (Hu et al., 2024a; J. et al., 2024; Chen et al., 2024). Consequently, they provide limited support for early-stage risk screening and proactive safety decision-making related to thermal runaway (Masalkovaitė et al., 2024; Zhang et al., 2024b; Wang et al., 2024; Yang et al., 2024). Recent studies have further expanded this understanding by examining thermal runaway behavior under more realistic service and abuse conditions, as well as by developing data-driven warning models under specific operating scenarios,

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such as high-rate charge/discharge, cycling-induced degradation, different states of charge, wrapping/insulation materials, and fire source-wall spacing (Zhou et al., 2025; Liu et al., 2026; Chen et al., 2025; Huang et al., 2025a, 2025b, 2025c). These works have deepened knowledge of battery failure evolution, propagation behavior, and safety characteristics under complex operating scenarios. Nevertheless, their emphasis remains primarily on mechanism analysis, propagation characteristics, and post-event hazard evaluation, rather than on pre-event consequence severity assessment.

Furthermore, from an engineering safety management perspective, predicting the occurrence of thermal runaway alone is often inadequate (Gu et al., 2025; Örs and Javani, 2025; Kumar and Kim, 2025). Even when thermal runaway is anticipated, its severity can vary considerably across different battery designs, capacities, geometries, and triggering conditions (Masalkovaitė et al., 2024; Ma et al., 2024; Meng et al., 2024; Liu et al., 2024). This variability directly influences the extent of damage, fire propagation potential, and the mitigation measures required. In contrast, pre-event estimation of thermal runaway severity would facilitate more efficient allocation of safety testing resources, early identification of high-risk cells, and informed decision-making during the battery design and validation stages (Hu et al., 2024b). However, developing such pre-event severity assessment methods remains a considerable challenge.

Physics-based and semi-empirical models have been explored to describe thermal runaway behavior. For example, J. et al. (2024). provided a comprehensive overview of thermal runaway modeling in lithium-ion batteries, encompassing abuse triggers, heat generation, propagation, and gas venting/combustion processes. They highlighted the role of validated models in improving battery safety assessment and design while reducing reliance on destructive experiments. Zhang et al. (2024b). developed an internal thermal runaway propagation model focusing on internal temperature evolution and configuration effects. It introduces triggering energy as a key metric for characterizing thermal runaway under thermal abuse, demonstrating that most energy during adiabatic runaway derives from internal exothermic reactions and that cell configuration significantly influences propagation behavior. Additionally, Xia et al. (2025). proposed a non-dimensional indicator, termed the Thermal Runaway Propagation Number, to quantitatively evaluate the conditions conducive to thermal runaway propagation in lithium-ion battery modules. Validated through experiments and simulations, this parameter provides clear criteria for suppressing propagation and guides the design of high-safety battery systems. However, their application to pre-event severity prediction is limited due to the highly coupled thermal, chemical, and mechanical processes involved. Accurate modeling typically necessitates detailed internal parameters, material properties, and reaction kinetics that are challenging to obtain during early design phases or before safety testing. Moreover, these models primarily aim to replicate specific failure scenarios rather than offering a generalized, deployable framework for severity screening across diverse battery populations.

Under these constraints, data-driven methodologies present a practical alternative for extracting latent safety-relevant patterns from heterogeneous and incomplete information available prior to thermal runaway (Ouyang et al., 2023; Ye et al., 2025). In recent years, machine learning techniques (Thelen et al., 2024; Wang et al., 2025b) have been increasingly employed in battery safety applications, such as thermal runaway detection and early warning systems based on temperature, voltage, or electrochemical signals. For instance, Masalkovaitė et al. (2024). developed a transfer learning approach to predict variations in thermal runaway heat output in lithium-ion cells, leveraging limited experimental data. This method markedly reduced the necessity for costly calorimetric measurements while preserving high prediction accuracy. Li et al. (2025). proposed a hybrid framework integrating experimental data, physics-based modeling, and deep learning techniques to predict thermal runaway temperatures in lithium-ion batteries subjected to extreme high-temperature shocks. The approach enabled

accurate and rapid predictions by combining experimental data with simulation-generated virtual datasets. Furthermore, Gajghate et al. (2025). developed a transformer-guided multimodal machine learning framework for predictive and causal assessment of thermal runaway in high-energy batteries. By integrating thermal, acoustic, and structural data with deep learning, causal modeling, and reinforcement learning, the framework facilitates accurate early warning, fault localization, and optimized sensor management in battery safety monitoring. Zhang et al. (2023). introduced a practical deep-learning framework for fault and anomaly detection in lithium-ion batteries, employing a dynamic autoencoder tailored to complex battery systems. Validated on extensive real-world electric vehicle datasets, this approach markedly enhanced detection performance and reduced battery fault inspection costs. Besides, Örs and Javani (2025). applied machine learning and deep learning algorithms to predict the onset of thermal runaway in lithium-ion batteries subjected to external heat and mechanical forces. By evaluating various models with experimental data, they demonstrated that decision tree-based models achieved the highest prediction accuracy, highlighting the potential of data-driven methods for thermal runaway forecasting. Although these studies illustrate the potential of data-driven approaches for safety monitoring, their primary focus remains on predicting the occurrence or initiation of thermal runaway. The assessment of the severity of thermal runaway events prior to occurrence, especially from a process safety perspective, remains largely unaddressed.

Therefore, despite the substantial progress in thermal runaway mechanism analysis, post-event consequence characterization, and abuse-condition-specific battery safety studies, an important gap remains in the literature. Existing studies rarely provide an interpretable framework for assessing how severe a thermal runaway event is likely to be before it occurs, using only information that is available at the design or pre-test stage. In particular, most available approaches either rely on post-event experimental measurements or focus on binary failure identification, rather than differentiating the severity levels of potential thermal runaway consequences in a form suitable for risk-oriented decision-making. This limitation constrains early-stage screening, conservative hazard identification, and efficient allocation of safety testing resources.

To address this gap, this study proposes a safety-oriented and interpretable machine learning framework for the pre-event prediction of lithium-ion battery thermal runaway severity based solely on structural and engineering metadata. A hazard-oriented Severity Index is first constructed to represent the combined consequence severity associated with heat release and material ejection, and thermal runaway events are then stratified into Low, Medium, and High severity levels. On this basis, a CatBoost-based classification model is developed to predict severity categories from pre-event battery metadata, and SHAP analysis is further applied to reveal the major features driving the prediction results. The proposed framework is intended to support proactive battery safety assessment, test prioritization, and risk-informed engineering decision-making.

2. Dataset description and analysis

2.1. Data source and feature construction

The workflow of this study is demonstrated in Fig. 1. The dataset used is derived from the Battery Failure Databank compiled by NASA and NREL (Finegan et al., 2024), which systematically collects and curates experimental reports on lithium-ion battery abuse testing and thermal runaway events over multiple years. This database has been widely used in previous studies to investigate battery failure behavior and safety characteristics, and provides a reliable and representative basis for engineering oriented analysis. The original database contains 365 thermal runaway test records. To reduce the confounding influence of charge-state variation, only samples tested at 100% state of charge

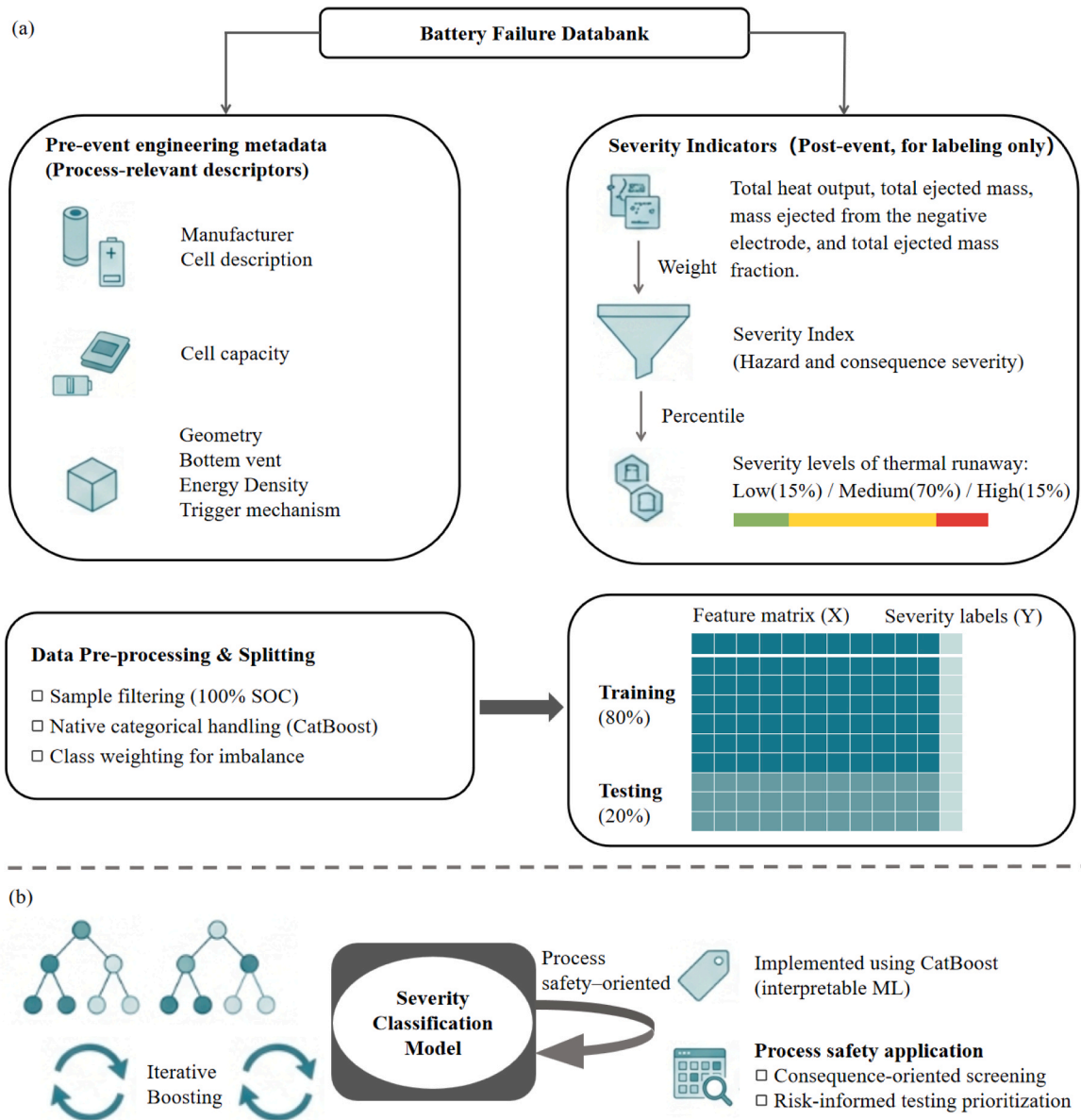


Fig. 1. Workflow of the proposed machine learning framework for pre-event prediction of lithium-ion battery thermal runaway severity. (a) Severity Index construction and dataset preparation. (b) Model training, interpretability analysis, and safety-oriented application.

(SOC) were retained in this study, resulting in a final dataset of 314 samples. The selected samples span multiple manufacturers, cell formats, trigger mechanisms, and vent configurations, with cell capacities ranging from 2.1 to 5.01 Ah, gravimetric energy densities from 143.78 to 279.88 Wh/kg, and volumetric energy densities from 315.68 to 767.81 Wh/L. A detailed summary of the dataset composition is provided in Table 1. Battery chemistry information is only partially available in the source database. Among the 314 selected samples, explicit cathode chemistry labels are reported for 48 samples, including NCA (27 samples), LiNiMnCoO₂ (15 samples), LiNiCoAl (2 samples), and one ambiguous LiNiMnCoO₂/NCA-related group (4 samples), whereas the remaining 266 samples do not have chemistry information explicitly recorded in the original database.

As this study is aimed at predicting thermal runaway severity prior to event occurrence, only battery structural and engineering metadata that are available before thermal runaway are considered as model inputs. Specifically, five categorical features and three numerical features are employed. The categorical features include manufacturer, cell description, geometry, whether cell has bottom vent, and trigger mechanism.

The numerical features include cell capacity, gravimetric energy density, and volumetric energy density. The selected input features were chosen according to three principles: pre-event availability, engineering relevance, and practical interpretability. The numerical features (cell capacity, gravimetric energy density, and volumetric energy density) reflect the energy storage level and energy concentration of the battery, which are directly related to the potential intensity of thermal runaway. The categorical features (manufacturer, cell description, geometry, bottom vent configuration, and trigger mechanism) describe structural design and initiation conditions that may influence heat accumulation, venting behavior, and material ejection during failure. Therefore, these variables provide a practical and safety-relevant metadata basis for pre-event severity prediction. In contrast, the output labels are determined by several post event physical quantities that directly characterize the hazard associated with thermal runaway. These quantities include the total heat output normalized by capacity, the total ejected mass, the mass ejected from the negative electrode, and the total ejected mass fraction. Together, these indicators capture both the energetic intensity and material ejection behavior of thermal runaway events, which are

Table 1

Summary of the battery samples included in this study.

Item	Value
Original database size	365 test records
Selection criterion	Pre-test SOC = 100%
Final sample size	314
Unique cell descriptions	30
Severity classes	Low (47), Medium (220), High (47)
Cell geometries	18650 (239), 21700 (48), 33600/D-cell (27)
Trigger mechanisms	Heater (Non-ISC) (156), Heater (ISC) (99), Nail (59)
Bottom vent	Yes (114), No (200)
Number of manufacturers	9
Cell capacity range	2.1–5.01 Ah
Gravimetric energy density range	143.78–279.88 Wh/kg
Volumetric energy density range	315.68–767.81 Wh/L
Chemistry information availability	Explicit cathode chemistry labels available for 48 of 314 samples
Reported chemistry categories	NCA (27), LiNiMnCoO ₂ (15), LiNiCoAl (2), ambiguous LiNiMnCoO ₂ /NCA-related group (4)
Samples without explicit chemistry labels	266

widely recognized as key contributors to thermal runaway severity. The purpose of this study is not to exhaustively include all possible descriptors, but to establish a lightweight and interpretable prediction framework based on metadata that are realistically available in engineering practice.

2.2. Construction of the thermal runaway severity index

2.2.1. Hazard-based rationale for severity definition

From a process safety perspective, thermal runaway severity cannot be adequately characterized by a single physical quantity (Sharp et al., 2022). Although total heat output is widely used to describe thermal runaway intensity, it reflects only the energetic aspect of the event and does not fully represent its hazard potential (Hu et al., 2024a). In practical safety scenarios, the consequences of thermal runaway depend not only on the amount of energy released but also on how that energy is discharged and interacts with the surrounding environment.

Thermal runaway is frequently accompanied by gas release and material ejection (Willstrand et al., 2025), which can significantly amplify hazard escalation. High-temperature particles, molten materials, and gas jets may directly damage adjacent components, promote fire propagation, and trigger secondary failures, especially in confined battery systems (Liu et al., 2025). Events with moderate heat release but aggressive ejection behavior may therefore pose comparable or even greater risks than events with higher energy output but limited mechanical impact. These considerations highlight the need for a severity definition that integrates both energetic and ejection-related characteristics.

2.2.2. Mathematical formulation and weighting strategy

Based on the above hazard-oriented rationale, a composite Severity Index is constructed by combining normalized indicators related to heat release and material ejection. Normalization ensures comparable numerical ranges and prevents scale effects from biasing the index formulation. The Severity Index is defined as a weighted summation of the selected indicators, including total heat output normalized by capacity, the total ejected mass, the mass ejected from the negative electrode, and the total ejected mass fraction.

The proposed Severity Index serves as an engineering-oriented surrogate for consequence severity, integrating both energetic intensity and material ejection characteristics relevant to process safety analysis. In the absence of a standardized severity metric and clear prior knowledge regarding the relative importance of individual hazard components, equal weighting is adopted in this study. This choice represents a

conservative and assumption-minimizing engineering strategy. Equal weighting avoids introducing subjective bias and ensures that both energetic and mechanical aspects of thermal runaway are considered with comparable importance. Such a formulation is well suited for safety screening applications, where underestimation of hazard is undesirable.

2.2.3. Percentile-based severity stratification and engineering interpretation

To enable practical severity classification, the continuous Severity Index (SI) is stratified into discrete severity levels using a percentile-based scheme. Absolute threshold values are avoided, as they are often sensitive to experimental conditions and data scaling in heterogeneous datasets. Instead, percentile-based stratification emphasizes relative hazard ranking among thermal runaway events.

Thermal runaway events are categorized into Low, Medium, and High severity levels according to the Severity Index distribution. The lowest 15% and highest 15% of SI values are assigned to Low and High, respectively, while the remaining 70% are labeled as Medium. This strategy avoids reliance on absolute thresholds and provides a stable, interpretable labeling scheme for subsequent classification. This relative classification also supports risk-oriented decision-making and facilitates the sensitivity analysis that follows. It should be emphasized that these three classes represent relative severity levels within the current dataset rather than universal hazard thresholds applicable to all battery systems. From an engineering perspective, the High severity class is intended to isolate the most hazardous thermal runaway scenarios characterized by strong combined heat release and material ejection, whereas the Medium class represents transitional cases and the Low class corresponds to comparatively limited consequences. This design improves the practical interpretability of the classification results and supports risk-oriented safety screening.

Fig. 2 illustrates the distribution of several selected input features across different thermal runaway severity levels. Distinct differences can be observed in the distributions of capacity and energy density across severity classes, with higher-capacity and higher-energy-density cells tending to exhibit more severe outcomes. This trend is consistent with their greater stored energy. Differences in geometry related categories further indicate the influence of cell structure on heat accumulation and failure evolution. In addition, the proportions associated with different trigger mechanisms vary notably across severity levels, highlighting the role of triggering conditions in shaping the intensity and consequences of thermal runaway events. These observations provide statistical evidence supporting the relevance of the selected metadata features for severity prediction.

3. Methodology

3.1. Problem formulation and learning framework

The proposed task is formulated as a supervised multi-class classification problem, aiming to predict the severity level of lithium-ion battery thermal runaway events before their occurrence. Each battery sample is represented by a set of easily accessible metadata features that can be obtained prior to thermal runaway, including manufacturer, cell description, geometry, whether cell has bottom vent, trigger mechanism, cell capacity, gravimetric energy density, and volumetric energy density. These features consist of both categorical and numerical variables, reflecting the structural and engineering characteristics of the battery.

The target variable is the thermal runaway severity level, categorized into three classes, namely Low, Medium, and High, based on the proposed Severity Index. From an engineering safety perspective, the primary objective of this classification task is not only to achieve high overall accuracy, but also to reliably identify high severity events, which correspond to thermal runaway scenarios with the greatest hazard potential. Therefore, special attention is paid to the prediction performance of the High severity class throughout model training and

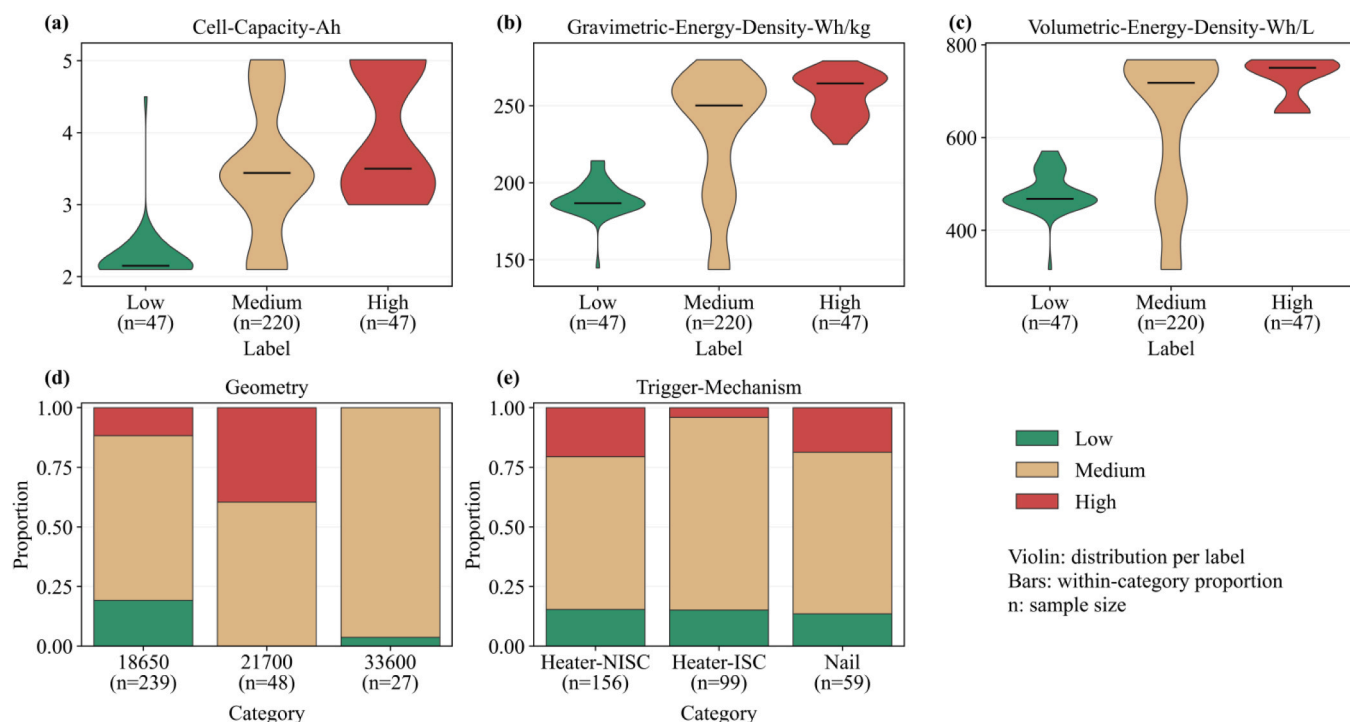


Fig. 2. Distribution of thermal runaway severity levels as a function of selected battery metadata features. (a) Cell capacity. (b) Gravimetric energy density. (c) Volumetric energy density. (d) Geometry. (e) Trigger mechanism.

evaluation.

3.2. Classification models

To investigate the suitability of different learning algorithms for the proposed severity prediction task, several widely used classification models are considered. Logistic Regression (LR) (Berkson, 1944) is adopted as a baseline model due to its simplicity and inherent interpretability. In addition, two tree-based ensemble methods, Random Forest (RF) (Breiman, 2001) and Extreme Gradient Boosting (XGBoost) (Chen and Guestrin, 2016), are included to represent commonly used nonlinear models capable of capturing complex feature interactions.

CatBoost (Dorogush et al., 2018) is a gradient boosting decision tree algorithm that constructs an ensemble of symmetric decision trees by sequentially minimizing classification loss. It incorporates ordered boosting and target-based encoding to effectively handle categorical features while mitigating target leakage. It is selected as the primary model in this study due to its strong compatibility with safety-oriented metadata and its ability to handle categorical dominance in a principled manner. Battery safety metadata are characterized by a high proportion of categorical variables, such as manufacturer, cell description, geometry, and trigger mechanism. These features encode critical design and operational information but are not naturally ordered or continuous. Conventional models often require extensive preprocessing, such as one-hot encoding or manual feature transformation, which can increase dimensionality, dilute informative structure, and introduce unintended bias, particularly when sample size is limited. CatBoost addresses these issues through native categorical feature handling and ordered boosting strategies, enabling effective learning directly from heterogeneous metadata without aggressive preprocessing.

The dominance of categorical features also poses challenges for many machine learning algorithms in safety applications. Categorical descriptors often act as proxies for latent design characteristics and safety margins that are difficult to quantify explicitly. Preserving their informational integrity is therefore essential for meaningful severity prediction. CatBoost's target-based encoding scheme allows categorical

features to be incorporated in a way that reflects their statistical relationship with the target variable while reducing the risk of target leakage. This property is particularly important in safety datasets, where spurious correlations can lead to overconfident predictions and unreliable risk assessment.

Another critical consideration in thermal runaway severity prediction is class imbalance. In safety-related datasets, high-severity events are inherently rare but carry disproportionate consequences. Traditional accuracy-oriented optimization may therefore favor majority classes while failing to identify the most hazardous cases. In this study, particular emphasis is placed on recall for the High severity class, as missed detection of high-risk thermal runaway events represents an unacceptable outcome from a process safety perspective. Classification models are thus evaluated not only by overall accuracy, but also by their ability to reliably capture high-severity cases. CatBoost provides flexible mechanisms for handling class imbalance, including class weighting and loss function adjustment, which enable conservative risk identification. By prioritizing recall for high-severity events, the model aligns with safety engineering principles that favor risk aversion over marginal gains in global accuracy. This alignment makes CatBoost particularly suitable for pre-event thermal runaway severity prediction, where the primary objective is early identification of potentially catastrophic outcomes rather than precise estimation of average behavior.

3.3. Training and validation strategy

The dataset is randomly split into training and testing sets using an 80/20 ratio. Model robustness is evaluated through five fold cross validation on the training data, reducing sensitivity to data partitioning. Early stopping and class weighting are applied during training to prevent overfitting and address class imbalance, with particular emphasis on accurately identifying High severity events.

3.4. Evaluation metrics

Model performance is evaluated using multiple complementary

metrics, including accuracy, Macro-f1 score, recall for the High severity class, and confusion matrices. Specifically, classification accuracy is reported to provide a general measure of prediction correctness. In addition, the Macro-f1 score is used to account for class imbalance by equally weighting the performance across all severity categories. Recall is specifically emphasized for the High severity class, as failing to identify high-risk thermal runaway events may lead to severe safety consequences in practical applications. Confusion matrices are further analyzed to provide detailed insights into class wise prediction behavior and misclassification patterns. Together, these evaluation metrics offer a comprehensive assessment of model performance from both statistical and engineering perspectives.

3.5. SHAP-based interpretability analysis

Interpretability is essential in safety-oriented applications, where predictive accuracy alone is insufficient to support engineering decision-making (Thelen et al., 2024; Wang et al., 2025b). For thermal runaway severity prediction based on battery metadata, it is necessary to understand how individual design and operational features influence risk classification. Such understanding enables validation of model behavior, assessment of physical plausibility, and extraction of actionable safety insights.

To address this need, SHapley Additive exPlanations (SHAP) analysis is applied to the trained CatBoost model (Lundberg and Lee, 2017). SHAP is an interpretable machine learning method grounded in cooperative game theory, which quantifies the contribution of each input feature to the model output. By decomposing predictions into additive feature attributions, SHAP enables both global interpretation of overall model behavior and class-specific analysis of prediction drivers.

SHAP is particularly suitable for the present study due to its compatibility with tree-based ensemble models and its ability to support multi-class interpretation. In particular, SHAP allows the key factors associated with High severity predictions to be explicitly identified and examined. This capability facilitates the extraction of engineering insights and supports the integration of data-driven prediction with process safety understanding in battery safety assessment.

4. Results

4.1. Comparison with baseline models

The performance of the proposed CatBoost classifier is compared with three baseline models, including Logistic Regression, Random Forest, and XGBoost. The comparison is conducted using accuracy,

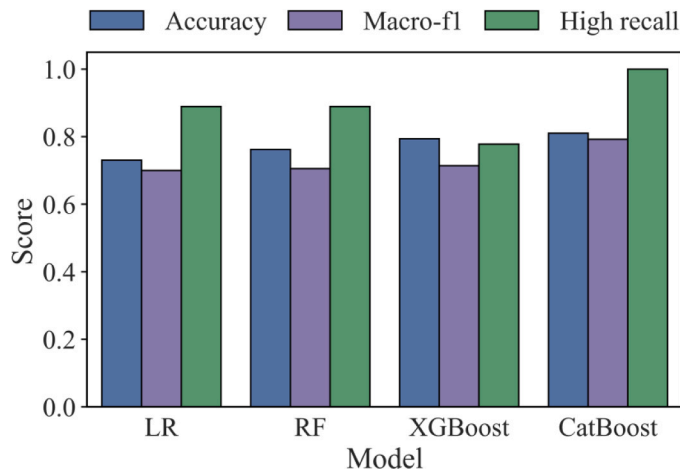


Fig. 3. Performance comparison of baseline classifiers using accuracy, Macro-f1 score, and High severity recall.

Macro-f1 score, and recall for the High severity class as evaluation metrics, as summarized in Fig. 3. Among all evaluated models, CatBoost achieves the best overall performance. It delivers the highest classification accuracy (0.81) and Macro-f1 score (0.79), indicating superior overall predictive capability across all severity levels. More importantly, CatBoost attains a recall of 1.000 for the High severity class, successfully identifying all high-risk thermal runaway events in the test set. In contrast, the baseline models exhibit lower recall values for the High severity category, indicating missed detections of critical events. While XGBoost shows competitive accuracy, its recall for High severity events is notably lower, suggesting a trade-off between overall performance and safety critical detection. Logistic Regression and Random Forest demonstrate comparatively stable behavior but are limited in their ability to simultaneously achieve high overall accuracy and reliable identification of high severity cases.

These results highlight the advantage of CatBoost in handling mixed categorical and numerical battery metadata under class imbalanced conditions. In particular, its ability to achieve zero missed detection for High severity thermal runaway events makes it well suited for safety oriented prediction tasks, where conservative risk identification is preferred over marginal gains in overall accuracy.

4.2. Performance of the CatBoost classifier

The performance of the CatBoost classifier was further evaluated using cross-validation and class-wise metrics. Fig. 4 illustrates the classification accuracy obtained across the five cross validation folds. Although performance variability is observed among different folds, the accuracy remains stable overall, with the mean value of 0.755 indicating consistent generalization capability.

To further examine model behavior across severity levels, class specific performance metrics are analyzed, as shown in Fig. 5. The model achieves high recall for the High severity class, indicating effective identification of the most hazardous thermal runaway events. This behavior aligns with the safety oriented objective of the proposed framework. For the Low and Medium severity classes, the classifier demonstrates balanced performance in terms of recall. The slightly reduced performance for the Medium class suggests a degree of overlap in feature characteristics between adjacent severity levels. Overall, these results indicate that CatBoost provides a favorable trade off between global classification performance and reliable detection of high severity events.

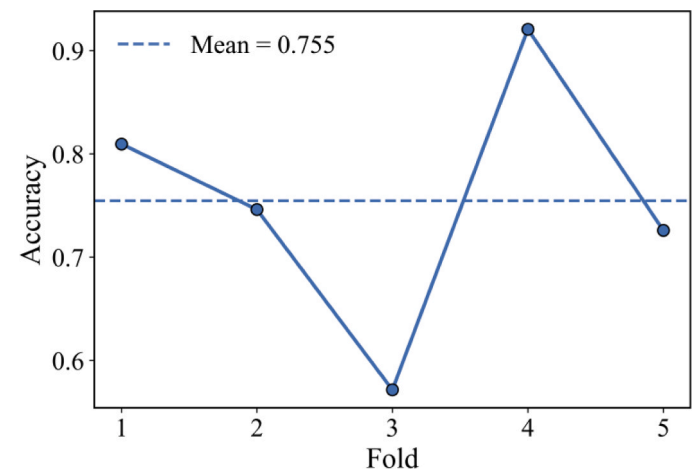


Fig. 4. Classification accuracy of the CatBoost model across five-fold cross-validation.

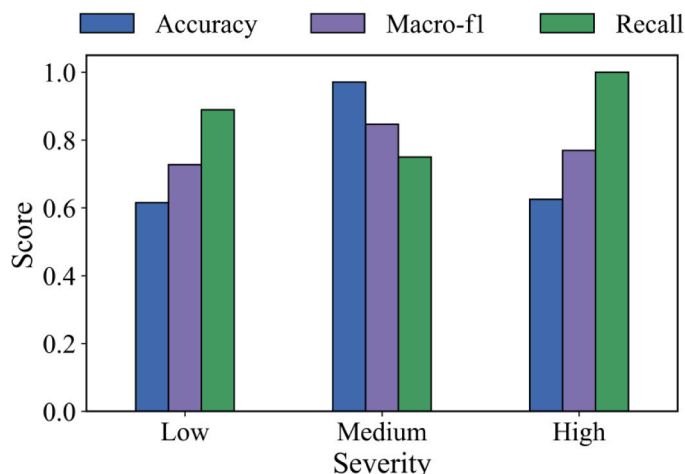


Fig. 5. Class-wise performance of the CatBoost classifier across different thermal runaway severity levels.

4.3. Misclassification analysis

To further examine the prediction behavior of the proposed model, a misclassification analysis is conducted based on the confusion patterns of predicted severity levels. Fig. 6(a) presents the distribution of true severity classes within each predicted category in terms of absolute sample counts, while Fig. 6(b) shows the corresponding class proportions. As shown, all samples with true High severity are correctly classified as High, indicating that the proposed model achieves zero missed detection for high-risk thermal runaway events. This result is particularly important from a safety perspective, as it demonstrates the model's ability to reliably identify the most hazardous cases. The majority of misclassifications occur in the Medium severity category. Specifically, Medium severity samples are occasionally predicted as either Low or High. This behavior is consistent with the underlying physical characteristics of thermal runaway severity, as Medium severity represents a transitional regime with overlapping features between low and high hazard levels. In contrast, misclassification between Low and High severity classes is rarely observed, suggesting that the model effectively separates extreme risk levels. Overall, the misclassification patterns indicate that the proposed approach prioritizes conservative identification of high severity events while maintaining reasonable discrimination among adjacent severity classes.

5. Explainability and physical insights

To further elucidate the physical implications of the SHAP-based feature importance results, the class-wise contributions of battery

metadata features are analyzed, as illustrated in Fig. 7. The SHAP distributions reveal that thermal runaway severity is primarily governed by the combined effects of stored energy magnitude, structural heat transfer characteristics, and failure initiation pathways.

Specifically, gravimetric energy density makes the largest overall contribution to severity prediction across all classes. This finding suggests that energy storage intensity is the dominant factor influencing thermal runaway consequences. From a thermochemical perspective, higher gravimetric energy density is associated with increased active material loading and a lower inert mass fraction. Together, these characteristics increase the volumetric heat generation rate during exothermic decomposition reactions (Hu et al., 2024a). This intensifies reaction self-acceleration and reduces the system's thermal buffering capacity, thereby amplifying temperature escalation, gas generation kinetics, and subsequent mechanical rupture. The strong SHAP contribution of gravimetric energy density therefore reflects its fundamental role in governing the upper bound of energy release and hazard escalation during thermal runaway. Cell capacity remains a critical determinant of severity outcomes by directly defining the total available electrochemical energy reservoir. Larger-capacity cells store greater quantities of reactive material, which prolongs reaction propagation duration and sustains high heat release rates once thermal runaway is initiated (Finegan et al., 2024). The coupling between total stored energy and reaction propagation length explains the strong contribution of capacity to High severity classification and is consistent with experimental observations linking large-format cells with intensified thermal and mechanical hazard manifestations. In addition, structural descriptors, including cell description and manufacturer, also demonstrate significant importance in severity prediction (Masalkovaitė et al., 2024). These categorical features encapsulate latent design variables such as electrode architecture, separator robustness, electrolyte composition, and safety margin design. Variations in these parameters influence internal heat generation homogeneity, gas venting stability, and structural integrity under rapid pressure accumulation. Their strong SHAP contributions suggest that design-level material selection and manufacturing consistency play a critical role in determining failure mode transitions and the extreme manifestation of thermal runaway severity. Cell geometry exerts a secondary but physically meaningful influence by modulating heat dissipation pathways and mechanical confinement conditions. Geometric parameters determine the surface-to-volume ratio, which directly controls the balance between heat generation and external heat rejection during runaway evolution (Sharp et al., 2022). Additionally, geometric constraints influence stress concentration and vent rupture dynamics, thereby affecting ejection intensity and directional hazard propagation. The SHAP results indicate that geometry acts as a spatial coupling factor linking internal energy release with external hazard manifestation. Moreover, trigger mechanism further contributes to severity differentiation by governing the

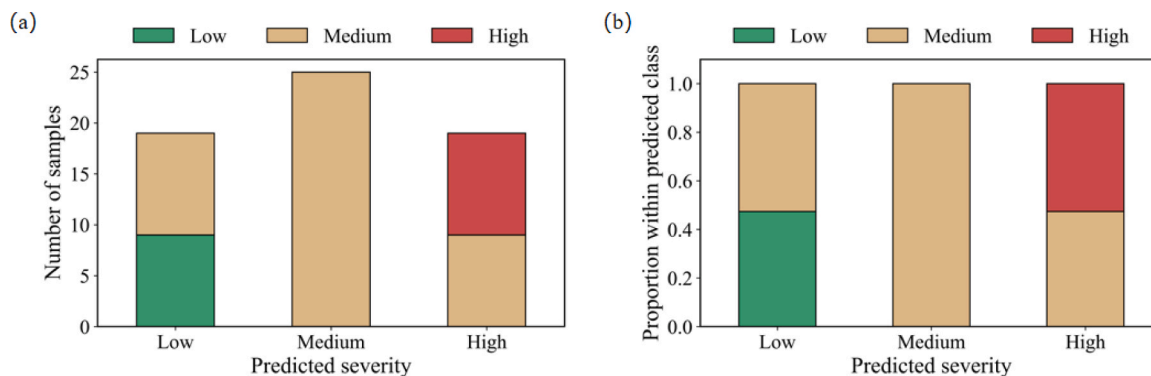


Fig. 6. Misclassification analysis of the CatBoost classifier showing the distribution of true severity levels within each predicted category in terms of (a) absolute sample counts and (b) class proportions.

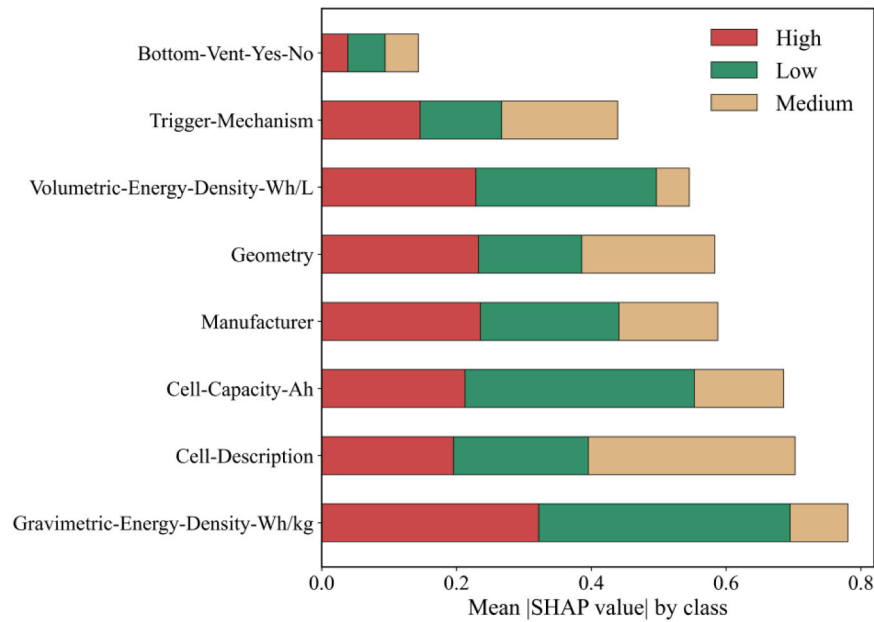


Fig. 7. SHAP-based feature importance analysis of the CatBoost model, showing the relative contributions of input metadata features to severity prediction.

spatial distribution and temporal rate of energy accumulation prior to runaway onset (Örs and Javani, 2025). Different triggering pathways, including thermal abuse, mechanical deformation, and internal short circuit, produce distinct temperature gradient profiles and localized reaction initiation patterns. These variations alter the synchronization of internal decomposition reactions and gas pressurization dynamics, which ultimately regulate the violence and propagation speed of thermal runaway events. The persistent contribution of trigger mechanism to High severity classification highlights the strong coupling between initiation pathway and consequence intensity. Also, volumetric energy density exhibits moderate influence by reflecting the degree of energy confinement within the cell structure. High volumetric energy density promotes localized heat accumulation and accelerates internal pressure buildup under limited thermal dissipation conditions, thereby facilitating rapid runaway escalation. In contrast, the relatively minor contribution of bottom vent structure suggests that pressure release features primarily function as mitigation mechanisms rather than primary determinants of severity magnitude (Liu et al., 2025). While venting can alleviate internal overpressure and redirect ejection flow, its effectiveness remains strongly dependent on the underlying energy release characteristics and structural stability of the cell.

Overall, the SHAP-based interpretation demonstrates that thermal runaway severity emerges from a multi-physics coupling between energy storage intensity, heat transfer limitation, structural integrity, and failure initiation dynamics. The alignment between data-driven feature importance and established thermal runaway mechanisms supports the physical credibility of the proposed prediction framework and provides quantitative guidance for energy-density optimization, structural safety design, and risk-oriented battery engineering.

6. Sensitivity analysis of percentile-based severity classification

In this study, the severity levels of thermal runaway events are determined using a percentile based classification of the Severity Index. This strategy avoids the use of absolute thresholds, which are often sensitive to experimental conditions and data scaling. It also ensures consistent severity labeling across heterogeneous datasets. However, the selection of percentile boundaries is not unique and may influence sample distribution, model training, and prediction performance. Therefore, a systematic sensitivity analysis of the percentile selection is

necessary. To evaluate the robustness of the proposed classification scheme, four percentile-based partitioning strategies, denoted as A, B, C, and D, are investigated. All schemes divide the Severity Index into Low, Medium, and High severity levels, but differ in the percentile boundaries and the resulting class proportions. The class proportions corresponding to the four candidate percentile schemes are summarized in Table 2, which provides the basis for the subsequent comparison of class distribution, model performance, and misclassification behavior. In scheme A, relatively narrow ranges are assigned to the Low (15%) and High (15%) severity classes, while the majority of samples are categorized as Medium (70%) severity, resulting in a broader intermediate severity interval. As shown in Fig. 8, the class distributions under different percentile strategies exhibit noticeable variations, particularly in the relative width of the Medium severity interval and the representation of extreme cases. As the percentile boundaries move closer to the median value, the proportions of Low and High severity samples increase, while the Medium severity class becomes narrower. Although such adjustments may enhance the statistical separation of extreme cases, they also reduce the representation of intermediate severity behaviors, which may increase class overlap.

Using each percentile scheme, CatBoost models with identical hyperparameters are trained and evaluated. The results shown in Fig. 9 indicate that the overall classification accuracy remains relatively stable across different percentile strategies, whereas the recall of the High severity class is more sensitive to the boundary design. However, the recall performance for the High severity class shows increased variability when more aggressive percentile boundaries are adopted. This suggests that an overly broad High severity interval may introduce more borderline cases into the extreme class, thereby weakening the model's ability to consistently isolate truly high-risk events.

Further analysis of the confusion matrices, as shown in Fig. 10, reveals that misclassifications predominantly occur between the Medium

Table 2
Proportion of different severity levels for all percentile schemes.

Scheme	Low	Medium	High
A	15%	70%	15%
B	20%	60%	20%
C	30%	40%	30%
D	33%	33%	34%

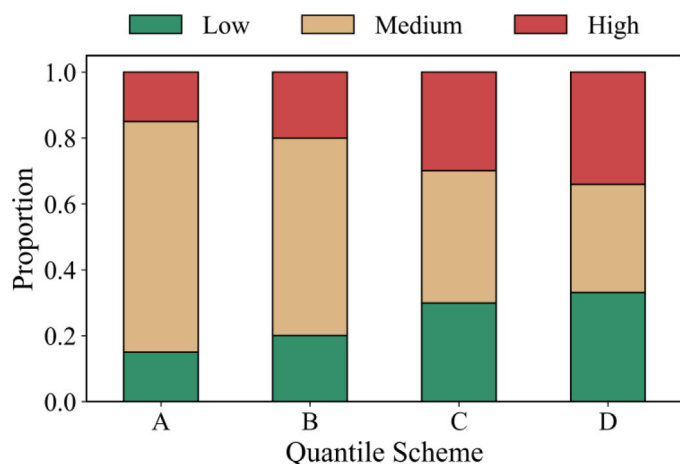


Fig. 8. Class distribution of Low, Medium, and High severity labels under different percentile schemes.

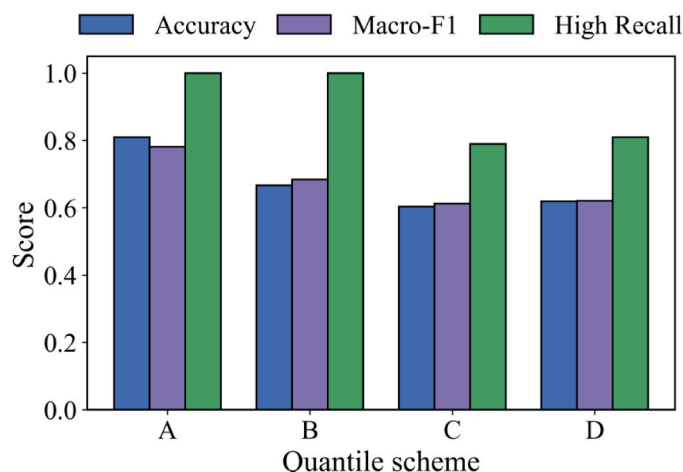


Fig. 9. Comparison of model performance metrics under different percentile schemes.

class and its adjacent severity levels. This behavior is physically reasonable, as Medium severity thermal runaway events often exhibit moderate heat release and ejection characteristics. These events naturally occupy a transitional region between low and high severity cases in the multi dimensional feature space. Excessive compression of the Medium severity range amplifies this overlap and increases classification ambiguity. The observed performance variations in Fig. 10 are not merely numerical fluctuations, but are closely related to how the percentile scheme reshapes class composition and separability. When the percentile boundaries move toward the median, the Medium severity interval becomes narrower, whereas the Low and High classes include more borderline samples. This reduces the distinctiveness of extreme classes and increases overlap between adjacent labels in the feature space. As a result, overall accuracy remains relatively stable, but the recall of the High severity class becomes less stable and misclassifications between the Medium class and its neighboring classes increase. From an engineering perspective, this indicates that an overly broad High severity definition may dilute the identification of truly hazardous events, whereas a broader Medium interval is better suited to accommodate transitional thermal runaway behaviors. This is why Scheme A offers the best balance between classification stability, hazard-oriented interpretation, and practical safety relevance.

From a physical perspective, scheme A provides a more interpretable and engineering relevant severity classification. By isolating the highest

Severity Index values into the High severity class, this strategy ensures that events with pronounced heat release and violent ejection behavior are clearly identified as high-risk. At the same time, the broader Medium severity interval accommodates the inherent variability of thermal runaway behaviors that do not exhibit extreme characteristics. Considering the combined evidence from class distribution, model performance, misclassification patterns, and physical interpretation of thermal runaway mechanisms, Strategy A is selected as the final percentile scheme in this study. Its advantage lies not only in performance stability, but also in preserving a hazard-oriented and interpretable definition of the High severity class. This strategy achieves a balance between reliable identification of high severity events and stable classification of intermediate cases. It therefore provides a robust and physically meaningful basis for subsequent severity prediction and engineering risk assessment.

7. Discussion

7.1. Comparison with existing thermal runaway severity assessment approaches

Thermal runaway severity assessment has traditionally relied on post-event experimental characterization, particularly calorimetric measurements and material ejection analysis. Techniques such as accelerating rate calorimetry, total heat release measurement, and ejected mass quantification provide detailed insights into the energetic and mechanical consequences of thermal runaway events. These approaches have played an essential role in understanding failure mechanisms and in establishing safety standards. However, they are inherently post-event methods, requiring the full development of thermal runaway and the availability of dedicated experimental facilities. As a result, their applicability is largely limited to retrospective analysis rather than proactive risk management.

In contrast, the framework proposed in this study addresses a fundamentally different problem: the prediction of thermal runaway severity prior to event occurrence. By relying exclusively on basic structural and engineering metadata that are available before abuse testing or deployment, the proposed approach shifts severity assessment from an experiment-intensive, post-event paradigm to a data-driven, pre-event risk evaluation task. This distinction is particularly important from a process safety perspective, where early hazard identification is a core principle for preventing catastrophic outcomes. Rather than replacing calorimetric or ejection measurements, the proposed method complements them by enabling early-stage screening and prioritization before costly and potentially hazardous experiments are conducted.

Another important difference lies in the conceptualization of thermal runaway outcomes. Many existing studies treat thermal runaway as a binary event, focusing on whether thermal runaway occurs or not under given abuse conditions. While binary classification is useful for identifying failure thresholds, it does not capture the substantial variability in consequence severity among different thermal runaway events. In practice, thermal runaway severity can range from limited heat release with minor venting to violent events involving large heat output and extensive material ejection. These differences have profound implications for system-level damage, fire propagation, and emergency response requirements. By introducing a Severity Index and stratifying thermal runaway events into Low, Medium, and High categories, the present work explicitly accounts for this variability and enables a more nuanced and decision-relevant representation of thermal runaway risk.

From a methodological perspective, recent data-driven studies on thermal runaway prediction have mainly adopted signal-driven algorithms for online warning or hazard evolution analysis, rather than metadata-based severity classification. For example, Vu et al. (2025). developed a real-time framework by combining an improved CUSUM algorithm with physics-informed Gaussian process regression GPR) to detect thermal runaway and estimate hazard severity using voltage and

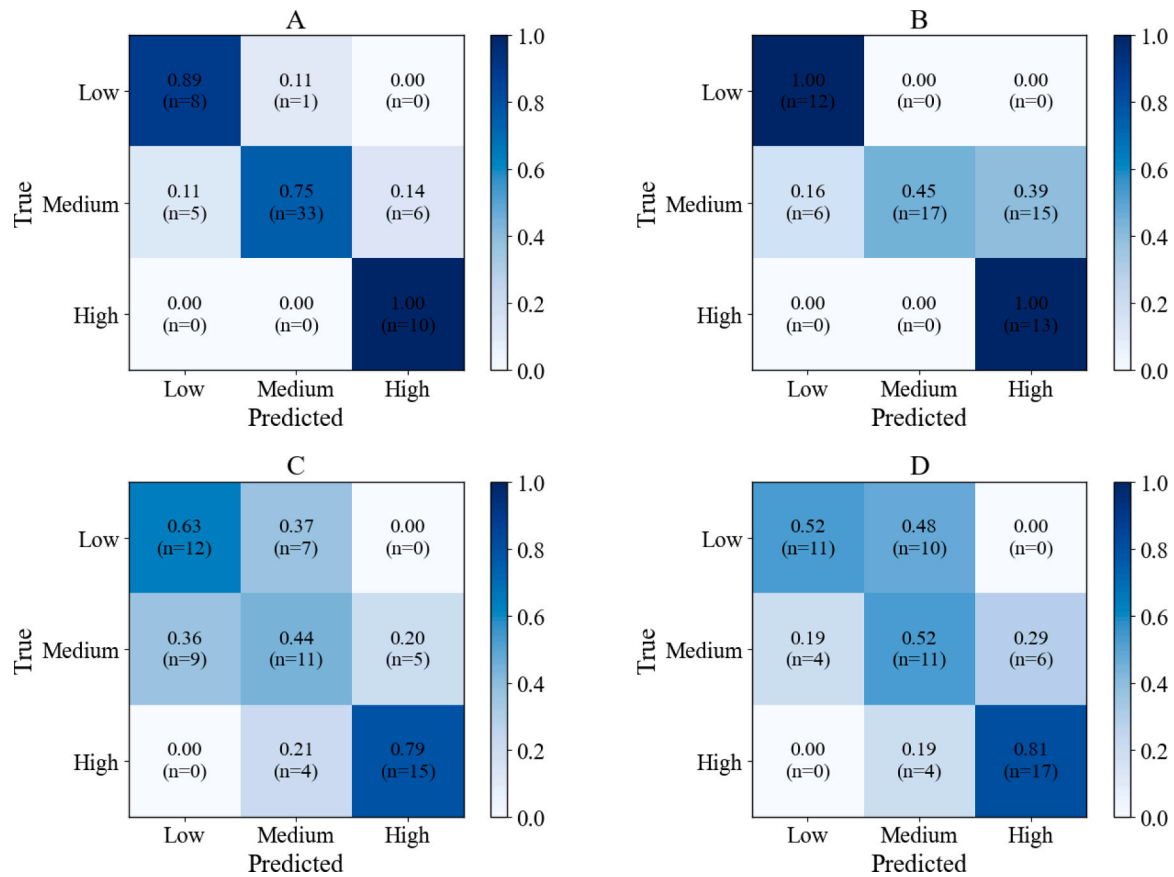


Fig. 10. Confusion matrices of the CatBoost model under different percentile schemes, with both class proportions and sample counts shown for each category.

temperature signals. Gu et al. (2025). proposed a state-of-safety SOS) warning framework based on strain, temperature rise, voltage, capacity, and power signals. Tam et al. (2025). developed an acoustic deep learning model for early-stage thermal runaway detection, while other recent studies have employed multiphysics-informed DeepONet models to predict thermal runaway evolution efficiently (Jeong et al., 2024). In contrast, the present study formulates thermal runaway severity prediction as a metadata-based multiclass classification problem and adopts CatBoost as the core learning algorithm. This choice is methodologically distinct from the above approaches. CatBoost is particularly suitable for heterogeneous tabular datasets containing both categorical and numerical variables, and it offers strong performance and robustness without requiring large-scale sequential sensing data. More importantly, the target of the present study is not online alarm generation or real-time hazard tracking, but the pre-event classification of severity levels using information available before testing or failure occurrence. Therefore, compared with recent signal-based data-driven methods, the proposed framework represents a complementary methodological route for safety-oriented battery evaluation at the design or pre-test stage.

Overall, compared with existing thermal runaway severity assessment approaches, the proposed framework introduces three key methodological features. First, it shifts severity assessment from post-event characterization and signal-driven online warning toward pre-event prediction based on readily available engineering metadata. Second, it represents thermal runaway consequence variability through a severity stratification scheme rather than binary failure judgment alone. Third, it formulates the problem as a CatBoost-based multiclass tabular learning task, which differs from recent regression-oriented, signal-fusion, deep learning, or physics-informed surrogate methods developed for online detection or dynamic evolution prediction. These characteristics make the proposed approach particularly suitable for early risk differentiation, conservative hazard identification, and pre-test screening in

process safety applications.

7.2. Engineering implications for safety testing and risk management

The proposed pre-event severity prediction framework has several important implications for battery safety testing and engineering risk management. One immediate application lies in test prioritization. Comprehensive thermal runaway experiments, especially those involving calorimetry and ejection measurements, are resource-intensive and carry inherent safety risks. In practice, it is often infeasible to subject all battery designs or variants to the same level of detailed testing. By providing an early estimate of thermal runaway severity based solely on metadata, the proposed method enables high-risk cells to be identified in advance and prioritized for more rigorous experimental evaluation. This risk-oriented testing strategy can significantly improve the efficiency and safety of laboratory workflows.

In addition to test prioritization, the framework supports early-stage screening during battery design and development. At early design stages, detailed experimental data are typically unavailable, while design choices related to capacity, geometry, and venting structure must still be made. The ability to assess potential thermal runaway severity using only basic design descriptors allows engineers to compare alternative designs from a safety perspective before committing to costly prototyping and testing. This capability is particularly valuable in the context of rapidly evolving battery technologies, where design iterations are frequent and safety considerations must be integrated early in the development cycle.

From a broader risk management perspective, the proposed severity stratification provides a more informative basis for decision-making than binary thermal runaway assessments. Differentiating between low, medium, and high severity thermal runaway scenarios enables more targeted mitigation strategies. For example, batteries predicted to

exhibit high severity thermal runaway behavior may require enhanced thermal barriers, increased spacing, or more robust venting and suppression systems at the module or pack level. Conversely, batteries associated with lower severity outcomes may allow for more flexible design trade-offs without compromising overall system safety. In this sense, the proposed framework supports a graded safety management approach, consistent with process safety principles that emphasize proportional risk control.

Resource allocation is another area where the proposed approach offers practical benefits. Safety testing capacity, engineering manpower, and mitigation resources are often limited. By enabling early identification of high-severity risks, the framework helps ensure that these resources are allocated where they are most needed. This is particularly relevant for large-scale applications such as electric vehicle fleets and grid-scale energy storage systems, where even a small number of high-severity failures can have disproportionate consequences. By enabling pre-event differentiation of thermal runaway severity, the proposed framework can be integrated into safety management workflows to support consequence-oriented process safety assessment and complement existing post-event experimental approaches.

8. Conclusion

This study developed an interpretable and safety-oriented framework for the pre-event prediction of thermal runaway severity in lithium-ion batteries using only structural and engineering metadata. To move beyond conventional post-event characterization, a hazard-oriented Severity Index was constructed to represent the combined consequence severity associated with heat release and material ejection, and was further used to stratify thermal runaway events into Low, Medium, and High severity levels.

Based on this formulation, a CatBoost classifier was established to predict severity categories from pre-event battery metadata. The model achieved strong overall classification performance and, importantly, demonstrated reliable identification of High severity events, indicating its suitability for conservative safety screening. SHAP-based analysis further revealed that gravimetric energy density, cell capacity, and structural descriptors are among the most influential factors governing severity prediction.

From an engineering perspective, the proposed framework provides a practical pathway for early-stage risk differentiation without relying on calorimetric or ejection measurements after failure. It can support battery safety screening, testing prioritization, and risk-informed design and management decisions in electric vehicles and energy storage systems. More broadly, this work offers a scalable route toward proactive and data-driven consequence assessment in lithium-ion battery safety engineering.

Code availability

The computational scripts developed for this work are publicly accessible via the GitHub repository (<https://github.com/HU-Qiqi/thermal-runaway-severity-prediction>).

CRedit authorship contribution statement

Qiqi Hu: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Minqiang Wu:** Writing – review & editing, Supervision, Conceptualization. **Chuxuan Wang:** Writing – review & editing. **Yun Da:** Writing – review & editing. **Hongda Du:** Writing – review & editing, Supervision, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT (OpenAI) in order to improve the language quality and clarity of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data used in this work were retrieved from the open-access Battery Failure Databank hosted by the National Renewable Energy Laboratory (<https://www.nrel.gov/transportation/battery-failure.html>). The source data supporting the conclusions of this study are provided with this article.

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